

Osprey — Lead Brief & Handover Meeting Agenda

Internal. For team leads. Read §1–3 before Monday (about ten minutes), bring §4 to the meeting with Aidan and Sean.

Meeting: Monday, with Aidan Astudillo and Sean Lee, who designed, built and raced Osprey.

This is probably the only time we get them in a room. They graduate, and most of what is in their heads was never written down. §4 is ordered so that if we only get through the first block, we still got the things that cannot be recovered any other way.

1. Where the project stands

Osprey is a 7 ft, 97 lb twin-motor electric catamaran, carbon hull laid up in-house. It planes and has run **30 mph** on the water. It races at the **ASNE/ONR Promoting Electric Propulsion** competition in Portsmouth VA, in the **Autonomy division** — a 2-mile course, four laps of an oval between two marks.

Three things happened at PEP26, all of them diagnosed by Aidan and Sean:

	WHAT	STATUS
Driveline	flex shaft escaped its collet mid-race, stern flooded	root cause known, redesign not done
Motor controller	one ESC failed on the bench, traced to a battery discharge-rating mismatch	packs already replaced; one ESC still needs replacing
Cooling	raw-water side never flowed reliably through the heat exchangers	never fixed, never properly tested

There is an autonomous system on the boat already. The thesis budgets 5 lb for it in the weight table and lists it as integrated into the electrical system alongside the RC gear — but then states plainly that it is *"outside the scope of this thesis"* [T §3.2]. So it exists, it is wired in, and **it was documented somewhere other than the thesis, or not at all.** Finding out which is the most important thing we get from Monday.

What the thesis does confirm: the RC system worked, giving stable wireless control and telemetry on the water, and early runs without payload reached 30 mph on plane [T §7.1]. Whether the autonomous system has ever actually steered the boat is not stated either way.

2. What has been added since

A manoeuvring simulation (Python) and an autopilot (an ArduPilot Lua script that the simulation flies) now exist at github.com/GrahamSmith2/Osprey. It is **not** a trajectory predictor — the hull coefficients are unknown to within $\pm 5\times$ and everything is built to survive that. Reference docs in the repo: `results/summary.md` (findings), `ASSUMPTIONS.md`, `docs/HANDOFF.md` (architecture and software gotchas), `docs/WORK_REMAINING.md` (full scope).

Four results that change what we should do:

The rudder should be sized on span, not area. Authority is the product $A_r \cdot CL_\alpha$. Growing area at fixed span grows the chord, which collapses the aspect ratio and cancels most of the gain: **30× the area buys 32% more authority.** Growing the span instead buys **5.3× for a ninth of the area increase.** The team has chosen **45 mm × 5 in: 2.7× the old blade's authority**, most of it from the extra depth.

At 50 mph, the hull unknowns decide whether the boat can turn. The nominal boat runs the 50 mph race cleanly with either blade. Across the plausible range of hull coefficients, **63% of boats with the old blade fail — spin out at the launch, cannot make a mark, or turn against the rudder — against 38% with the new one.** Dropping to 20–40 mph only brings that to ~29%. What closes it is measuring the hull, not slowing down or retuning.

The powertrain is correctly sized. The boat just is not using it. Hull resistance implies ~10 kW effective at 80 mph, so ~16 kW electrical at the 63% chain efficiency their own log implies, which is **180 A per motor.** The Hydra Cobra 5 HV is rated **180 A continuous:** $180 \times 44.4 \times 2 = 16.0$ kW against a 16.0 kW requirement. The hardware matches the design point almost exactly.

What is 4× short is the **actual current draw** — 45 A per motor where 180 A is available. That is a propeller and loading problem, not an installed-power problem, and no amount of new electronics fixes it.

Rule 21 requires a kill on GPS loss. The craft must stop when navigation data is missing or corrupted, and judges watch it demonstrated. Our Lua autopilot disarms 500 ms after the fix is lost, in simulation. Whether the existing autonomous system on the boat does this is unknown.

3. Decisions we owe ourselves

#	DECISION	BLOCKED ON
D1	What speed are we actually building for?	Decided: 50 mph cruise [TEAM, Sep 2026]. Sets the ESC, wiring, battery and cooling load: ~7.5 kW into the water, ~135 A per motor at the logged chain efficiency. The thesis already floats LMT 7065 motors, which would need a new front support plate.
D2	Driveline retention scheme	Nothing. Someone needs to own it.
D3	Rudder size and stock position	Size decided: 45 mm chord × 5 in submerged. Stock position is one measurement (§4, A2) and decides whether the servo is adequate.
D4	Steering allocation architecture	Decided: motors steer below 10 mph, rudder above. Implemented in the Lua autopilot, which drives the outputs itself.
D5	Are we entering PEP27 in Autonomy again?	Determines whether autonomy is the priority or a stretch goal

The open question behind D1: a boat that reliably finishes scores better than a fast one that does not — completing the distance is worth 40 points against 20 for winning — and the simulation says 50 mph is where the hull unknowns matter most (§2). Confirm the choice once the on-water hull data is in.

4. Questions for Aidan and Sean

Block A — things only they know

If we run out of time, we must have got through this block.

#	QUESTION	WHY
A1	What exactly wore on the collet, and do you still have the failed part?	The redesign depends on the wear mode. A photo or the part itself is worth more than the write-up.
A2	Where does the rudder stock sit on the blade, as a fraction of chord?	Decides whether the existing servo is adequate. For the new 45 mm blade at 50 mph the answer swings the requirement from ~0 to ~103 kg-cm (stall is 74). We were going to measure it; you may just know.
A3	What is the rudder tiller arm radius?	Scales servo torque linearly. Never recorded.
A4	Why do you think the raw-water side never flowed? Air lock, pickup geometry, insufficient dynamic pressure, blockage?	Determines whether we add pumps or redesign the pickups.
A5	What is your read on 45 A when you expected 100? Prop pitch, diameter, slip, cavitation?	This is the single biggest open question on the boat. The ESC is rated 180 A and the boat used 25% of it, so speed is being left on the table somewhere between motor and water.
A5b	Did anyone ever pull the Castle Link RPM and power logs?	The thesis says Castle Link exposes exactly the RPM and power data needed for propeller selection, and that they never got to use it. If those logs exist they settle A5 outright. If not, it is the first test we should run.
A6	Was there ever a problem with the 12 V battery, or is it just old?	The work list says "new 12 V battery" but the thesis reports no fault.
A7	What did you try on cooling that did not work?	Stops us repeating it.
A8	How much of the boat is one-off? Which parts would be hard to replace if we broke them?	Risk register for on-water testing.

Block B — files and hardware to walk away with

Ask for these explicitly. Get them on a drive in the room if possible.

#	ASK
B1	CAD — hull, molds, driveline, steering assembly. Native files, not just STEP.
B2	The MATLAB scripts from Appendix A.1 — hull sizing, <code>aeroForce</code> , <code>calcWettedAreaIterative</code> . We reimplemented the hull model independently and want to cross-check.
B3	Raw data logs — the 30 mph run, ESC ripple logs, any temperature data. Specifically the Castle Link logs : the thesis notes Castle Link exposes temperature, RPM and power output, and that they never got to use it properly. Those logs would go a long way toward answering A5.

ASK

B3b	Anything from whoever built the autonomous system — see Block D.
B4	Are the hull molds still usable, and where are they?
B5	Spare parts inventory — shafts, collets, props, connectors. What is on the shelf?
B6	Anything they'd want back, and permission to keep their thesis in our public repo (currently there with attribution; better to have it explicitly).

Block C — competition and race day

#	QUESTION	WHY
C1	Confirm the course: two marks, quarter mile apart, oval, four laps. Is a lap really half a mile?	Marks 0.25 mi apart puts 0.5 mi of straight in a lap <i>before</i> turning, so a lap is 0.54–0.58 mi. The two figures cannot both be exact.
C2	How is distance actually scored — GPS track, lap counting, or judges?	Changes whether tight rounding helps or hurts.
C3	What did the judges actually care about at inspection?	Kill switch, buoyancy, tow points, C-rating. Where did teams lose points?
C4	How did the 5-minute launch go? What slowed you down?	It is a hard gate.
C5	What surprised you about race day?	Open-ended and usually the most valuable question in the room.
C6	What did the boats that beat you do differently?	

Block D — the autonomous system, which the thesis deliberately excludes

The thesis budgets 5 lb for an "Autonomous System" and lists it as integrated into the electrical system, then says it is outside its scope. Everything below follows from that one sentence, and D1–D3 are close to Block A in priority.

#	QUESTION	WHY
D1	What actually is the autonomous system? Which flight controller, GPS, compass, telemetry radio, companion computer?	5 lb is a substantial amount of hardware. We may already own most of what we need.
D2	Who built it, and where is it documented? Another student, another thesis, a repo, or nothing?	If there is a second document we should be reading it before Monday, not rediscovering it.
D3	Has it ever driven the boat? Even a straight line, even a single waypoint.	Decides whether next year is "integrate and tune" or "start from scratch".
D4	Is it wired to the steering servo and throttle now, and what happens on RC/autonomous handover?	Handover behaviour is a safety item and a competition requirement.

#	QUESTION	WHY
D5	What is the RC failsafe today? What does the boat do on signal loss?	Rule 21 needs a kill on bad navigation data; the RC failsafe is the obvious place to hang it.
D6	How does the boat behave at low speed? Does it steer at all below planing, and what happens coming off the plane?	
D7	Does it track straight hands-off, or is there a turning bias?	
D8	What does it do in a hard turn — hook, slide, lift a sponson?	

D6–D8 matter because the simulation is guessing at exactly these behaviours within a factor of five, and they have stood on the bank watching the boat do them.

Block E — judgement, not data

Worth reserving ten minutes for. Ask these even if the meeting is running long.

- ****If you had another year on this boat, what is the first thing you would**

change?**

- **What did you underestimate most?**
- **What is the biggest risk you think we are not seeing?**
- **Is 80 mph realistic with this powertrain, or was that always aspirational?**

(The ESC is rated for it almost exactly. The gap is current draw, which points at the prop. Their view settles D1.)

- ****You mention the LMT 7065 as a future motor. How seriously did you look at**

it, and what made you stop short?**

- **Would you do the autonomy division again, or race a different class?**

Already answered — do not spend meeting time on these

The thesis §7.2 solves these. Listed so nobody rediscovers them the hard way.

- **Bending the brass stuffing tubes.** Kinking is easy and hard to repair, and

brass springback beats a jig on its own. Their fix: anneal the tubes in a filament dryer at 85 °C while held in the jig. Print the jigs in **ASA**, not PLA — PLA's glass transition is 60–65 °C and the jigs warp at annealing temperature.

- **Cooling, the direction of the fix.** Fit pumps to pull raw water rather than

relying on hydrodynamic pressure at speed. A4 is about *why* it failed, not what to do instead.

- **Tests they know were never done:** range testing (distance, time, speed and

power draw for a given pack), CG sensitivity, and sustained thermal. Worth confirming none of these happened quietly after the thesis was submitted.

5. Running the meeting

- **Record it**, with their permission. Half of Block E will be worth re-listening to.
- **One person on questions, one taking notes.** Do not have everyone interrogating at once.
- **Get Block B on a drive before anyone leaves the room.** Files are the thing that quietly disappears after graduation.
- **Bring the boat if it is accessible.** A2, A3 and A8 get answered in thirty seconds standing next to it, and they will point at things they would never think to write down.
- Circulate answers back into `docs/WORK_REMAINING.md` afterwards — anything currently marked **[OPEN]** should be resolvable from this meeting.